Caoliwen Wang

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Research Interests

Rooted in computer graphics, scientific computing, and generative AI, my research develops **physically grounded, design-ready, and scalable generative models** for physical world simulation, video generation, 3D shape generation, and scientific discovery.

Education

University of British Columbia Sep 2025 - Present
Ph.D. in Computer Science
Advisor: Prof. Peter Yichen Chen

University of Science and Technology of China Sep 2021 - Jun 2025
B.S. in Computational Mathematics
Advisor: Prof. Ligang Liu

Research Experience

Scalable Physical Simulation Apr 2025 - Present

- **WorldParticle**: a unified world model that represents diverse physical phenomena as Lagrangian particles and learns their dynamics with a single Transformer-based backbone, enabling physics-grounded video generation and 3D dynamic generation.

Physically Plausible 3D Generation Apr 2025 - Oct 2025

- **Gauss–Seidel Projection**: a method that enforces physical validity (non-penetration, structural plausibility) on generated 3D protein configurations and improves the inference efficiency of protein structure prediction foundation models.

Design-Driven 3D Shape Generation Jan 2024 - Jun 2025

- **Neural Shadow Art**: an implicit neural 3D representation that facilitates geometry generation from multi-view binary shadow targets, with angular adjustment and volume/geometry optimization.
- **Shape from Semantics**: a 3D generation framework for creating semantically consistent shapes whose multi-view renderings align with user-specified prompts, extending the inverse-modeling framework to a semantics-conditioned 3D generation problem.

Publications & Preprints

[1] **Caoliwen Wang***, Minghao Guo*†, Siyuan Chen*, Heng Zhang, Mengdi Wang, Xingyu Ni, Hanson Sun, Kunyi Wang, Zherong Pan, Kui Wu, Lingjie Liu, Yin Yang, Chenfanfu Jiang, Taku Komura, Wojciech Matusik, and Peter Yichen Chen†. WorldParticle: Unified World Simulation of Lagrangian Particle Dynamics via Transformer. *Preprint, 2026.*

[2] Siyuan Chen*, Minghao Guo*†, **Caoliwen Wang**, Anka He Chen, Yikun Zhang, Jingjing Chai, Yin Yang, Wojciech Matusik, and Peter Yichen Chen†. Physically Valid Biomolecular Interaction Modeling with Gauss-Seidel Projection. *International Conference on Learning Representations (ICLR), 2026.*

[3] Liangchen Li, **Caoliwen Wang**, Yuqi Zhou, Bailin Deng, and Juyong Zhang†. Shape from Semantics: 3D Shape Generation from Multi-View Semantics. *Preprint, 2025.*

[4] **Caoliwen Wang**, Bailin Deng, and Juyong Zhang†. Neural Shadow Art. *Pacific Graphics (Conference Track), 2025.*

Awards

University and Provincial Outstanding Graduate Award 2025 (USTC)

Reviewing

ACM SIGGRAPH Asia, Pacific Graphics